COD Risk Guard

Founder decision memo | Associate Software Engineer take-home | 30 October 2026

To: Founder and Operations / Marketing leads
Subject: A cautious, customer-preserving policy for COD RTO risk

Executive recommendation

Pilot a four-action, risk-based workflow rather than automatically cancelling high-risk COD orders. Ship low-risk orders, request WhatsApp confirmation for moderate risk, offer a small prepaid discount for higher risk, and route very-high-risk orders to human review. Keep automatic cancellation disabled. This balances operational cost reduction with the risk of rejecting genuine first-time customers.

Why this balance matters

The business brief reports that 55% of orders are COD and around 28% of COD orders become RTO. Each RTO is estimated to cost ■169.40: ■65 forward shipping + ■65 return shipping + ■25 packaging/handling + ■14.40 expected damage (8% × ■180). The brief also reports a 4% prepaid RTO rate and Meta acquisition costs of ■300+. Operations therefore has an avoidable-cost problem, while Marketing has a legitimate customer-acquisition and trust concern.

Evidence from the supplied historical data

Finding	Observed result	Implication
Completed outcomes	1,987 orders; 565 RTO; 28.43% RTO rate (Apr–Sep 2026)	The sample broadly reflects the brief's ~28% COD RTO assumption.
State variation	Bihar 53.79% (145 orders); Uttar Pradesh 44.10% (195); Madhya Pradesh 38.10% (126)	Use geography as a monitoring signal, not a blanket block; group rates are not causal proof.
Acquisition source	Meta ads 31.24% (765); direct 22.92% (253)	Review acquisition cohorts, but do not reject Meta customers solely on source.
Prior RTO history	0 prior RTO: 27.15% (1,845); 1: 35.19% (108); 2: 76.47% (34)	Prior RTO count may help prioritise verification; the two-RTO group is small.

September out-of-time backtest

A Logistic Regression model was trained on 1,658 April–August orders and evaluated on 329 September orders. September had 86 RTOs (26.14%). Ranking performance was ROC-AUC 0.6822 and PR-AUC 0.4051. At the 0.18 classification threshold selected earlier on validation data, precision was 31.56%, recall 89.53%, and F1 46.67% (77 true positives, 167 false positives, 9 false negatives, 76 true negatives). At 0.50, recall fell to 10.47%. The lower threshold catches more RTOs but flags many delivered orders; it is not a cancellation rule.

The four-action policy assigned September orders as follows:

Action	Orders	Actual RTO rate
Ship as-is	45	11.11% (5/45)
WhatsApp confirmation	181	21.55% (39/181)
Prepaid nudge	96	41.67% (40/96)
Hold for manual review	7	28.57% (2/7)

The prepaid-nudge bucket had a higher observed RTO rate than the ship-as-is bucket, making it a sensible segment for a controlled test. However, 284 of 329 orders (86.3%) were assigned an intervention or manual review. That is a significant operational workload and should be monitored.

Financial trade-off and proposed policy

The assignment's aggregate rates imply an expected RTO-cost reduction of 24 percentage points when a COD order converts to prepaid. At ■169.40 expected RTO cost, that is approximately ■40.66 avoided cost per successful conversion before discount. With a ■20 discount, the simple expected benefit is ■20.66 per converted order, before payment fees and other effects.

For September, the prototype reports ■6,715.26 estimated RTO savings, ■384 expected discount cost, ■0 cancellation loss, and ■6,331.26 estimated net impact. These are scenario outputs, not realised savings: they assume 50% intervention effectiveness and 20% prepaid conversion, and use model risk scores that have not been calibrated. They must not be presented as actual profit or proven impact.

The ■300+ Meta acquisition cost should be treated carefully. For an order already acquired, CAC is generally a sunk acquisition expense; do not add ■300 to every cancellation as if it were a new avoidable cost. The downside of cancelling a genuine order is lost contribution and possible repeat-purchase/trust effects. Gross profit at the brief's AOV and margin is ■402.38 (■649 × 62%), but this is not necessarily the exact contribution lost on every cancelled order.

Recommended initial rules

• Below 15% estimated risk: ship as-is.
• 15% to below 30%: send WhatsApp confirmation; YES proceeds, NO cancels at the customer's request, unclear replies trigger clarification, and no reply after 24 hours goes to manual review.
• 30% to below 55%: offer ■20 prepaid discount; if declined, continue with COD confirmation rather than automatically cancelling.
• 55% and above: hold for human review. Do not automatically cancel based only on model score.

Failure modes and monitoring

Monitor RTO rate, delivered-order false positives, confirmation response rate, prepaid conversion, cancellation/hold rates, discount cost, RTO cost per order, and customer complaints. Break results down by state, courier and acquisition source, but avoid blanket pincode/state blocks without sufficient volume and validation. Address-quality flags need care: short or unusual text can be legitimate Hinglish/local directions. The prototype does not integrate a live WhatsApp API, payment gateway, or LLM-based address interpretation. Model probabilities are not calibrated, and performance may drift across cohorts.

Two-week pilot plan

Week 1 — Shadow mode: Score incoming COD orders without changing dispatch. Review high-risk and false-positive cases with Operations and Customer Support. Confirm message wording, response handling and escalation ownership. Do not block customers or pincodes automatically.

Week 2 — Limited intervention: Run a monitored pilot for eligible confirmation and prepaid-nudge orders, ideally with a comparison/control group. Track response and conversion rates, delivered/RTO outcomes, discount spend, operational workload and complaints. Compare net contribution rather than counting every flagged order as a saved RTO. Adjust thresholds only after measured results.

Decision requested

Approve a cautious two-week pilot with human review for the highest-risk band, a ■20 prepaid offer as a test, no automatic cancellation, and daily monitoring. Expand only if measured RTO-cost reduction exceeds discount and operational costs without unacceptable genuine-customer friction.

Method note: Historical group differences are associations, not causal effects. September backtest is out-of-time but small (329 orders). Scenario financial estimates depend on unverified assumptions; validate with controlled outcomes before claiming savings.